

03 – Dataverse Data Model (tables, columns, sample data)

All tables are created in solution **TVRMS** with prefix `tvr_`. How to create a table: **Solution → New → Table → Table**, give the name, then add columns.

1. Big picture – how all the tables are linked (Entity Relationship)

This section explains **how the tables are linked**: 2 short lessons (1.1 and 1.2), then **2 pictures that together show all 26 links** (1.3), a **list of the 26 links** with a simple meaning and an example (1.4), and a **third picture with real data** (1.5). Each picture has its own landscape page, so the text is big.

1.1 First learn 4 simple words

Word	Simple meaning	Example in this project
Table	A list of records – like one sheet in Excel	TV Request , Department
Row (record)	One line in a table	The request TV-2026-00015
Lookup column	A column whose value is one row of another table – like picking from a dropdown	On the request, the column <i>Department</i> holds "Process Engineering"
Relationship (link)	The link that a lookup column creates between two tables	TV Request → Department

1.2 How to read the pictures

What you see	Meaning
A short bar and the letter 1 at the end of a line	The ONE side (the "parent" table)
A crow's foot (three short lines) and the letter N	The MANY side (the "child" table): zero, one or many rows
Several branches joined to one trunk	The single bar + 1 on the trunk is the ONE end of every branch – read each branch as its own sentence
The arrow → between two table names (in the text and in the cards)	"points to" : <i>TV Request → Department</i> = "a request points to one department". It always goes from the table that holds the lookup column (N) to the ONE table
The colour of a line	The colour of the table on the ONE side: blue = master data, purple = TV Request, grey = User, green = Feasibility Option (only the <i>Selected Option</i> link)
A dashed loop under the TV Request card	The self link for <i>Repeat</i> requests (empty for <i>New</i> requests)

Read every line as a sentence. Example: *Department 1 — N TV Request* means "**one** department can have **many** requests, but **each** request belongs to **only one** department".

Golden rule: the lookup column is always stored on the **N side** (the child). So the *Department* column is created on the **TV Request** table – not on the Department table.

Box colours: **blue** = master data · **purple** = main table (TV Request) · **green** = child tables · **grey** (dashed) = User, the people.

1.3 The three pictures – what each one shows

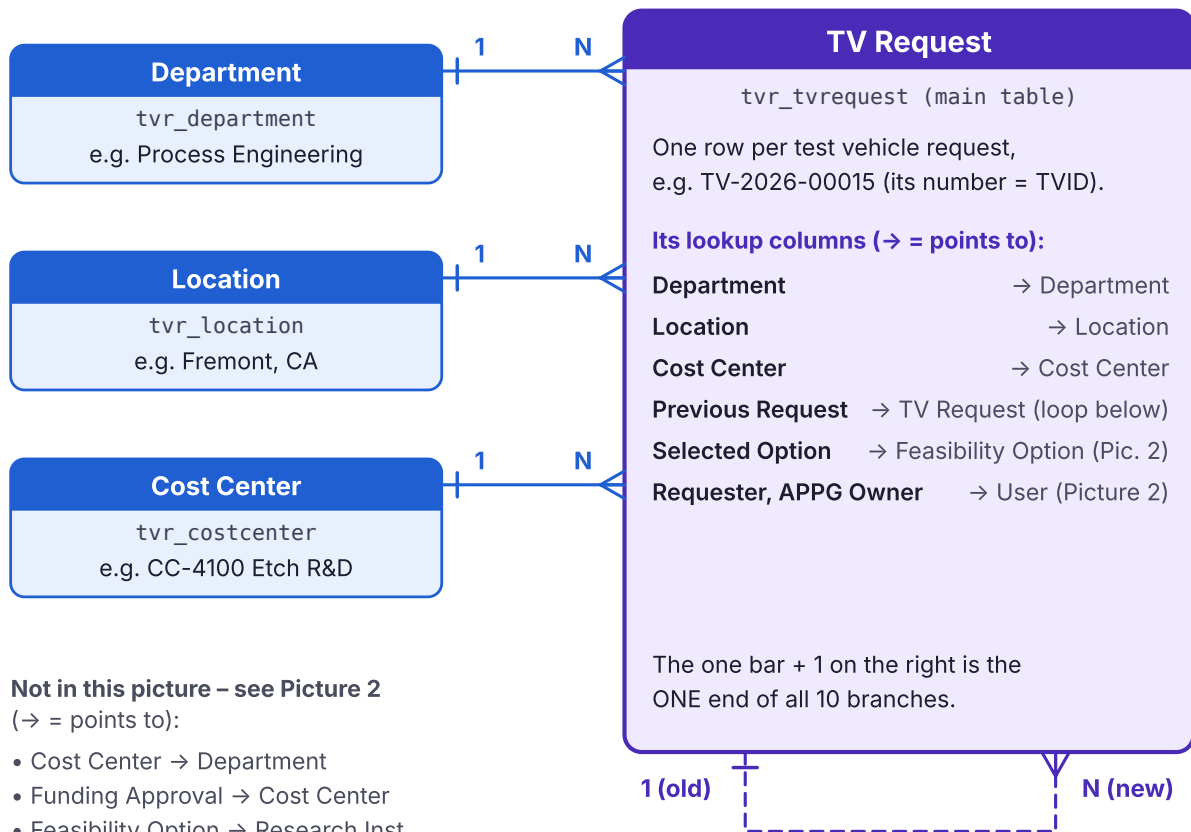
- **Picture 1 – TV Request, its 3 master-data parents, its 10 child tables and the self link (14 links).** *TV Request* is in the middle. On the left are its 3 master-data parents (Department, Location, Cost Center): each request points to one row in each. On the right are its **10 child tables**: every child table has its own branch, because every child table has a lookup column *TV Request*. The small tag on each child (Phase 3, Phase 5, Audit ...) shows when its rows are created. The dashed loop under the card is the self link for *Repeat* requests.
- **Picture 2 – the other 12 links.** Part A shows the 4 table links that are not in Picture 1: Cost Center → Department, Funding Approval → Cost Center, Feasibility Option → Research Institution, and TV Request → Feasibility Option (the lookup *Selected Option*). Part B shows the 8 "person" columns (for example *Requester* and *APPG Owner* on TV Request): all of them point to the built-in **User** table.
- **Picture 3 – real data** (in 1.5): one real request and the real rows linked to it.

Not linked to anything, so not in the pictures: **App Setting** and **Email Template** (the flows read them by key or code) and the optional **Status Transition** table (section 4).

Picture 1 – TV Request, its 3 master-data parents, its 10 child tables and the self link (14 links)

Every line is one lookup column. Bar + "1" = the ONE side. Crow's foot + "N" = the MANY side (0, 1 or many rows). → = points to.

PARENT TABLES – master data (ONE side)

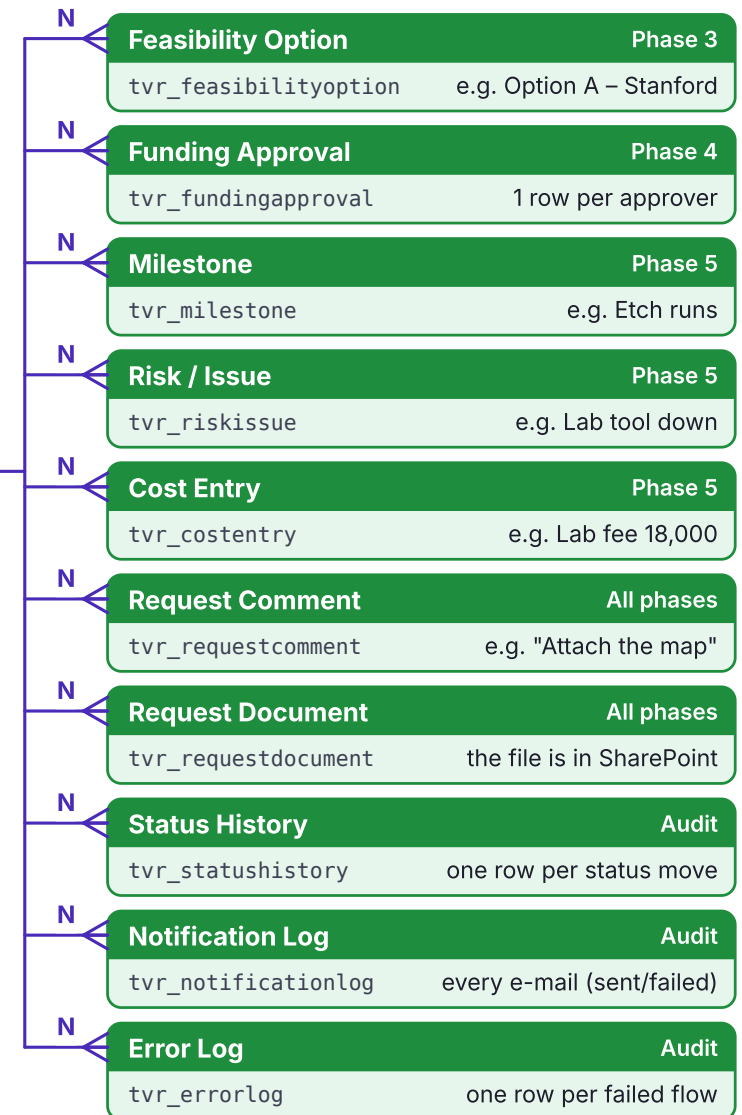


Not in this picture – see Picture 2

(→ = points to):

- Cost Center → Department
- Funding Approval → Cost Center
- Feasibility Option → Research Inst.
- TV Request → Feasibility Option (lookup "Selected Option")
- 8 person columns → User

CHILD TABLES – each has a lookup column "TV Request" (MANY side)

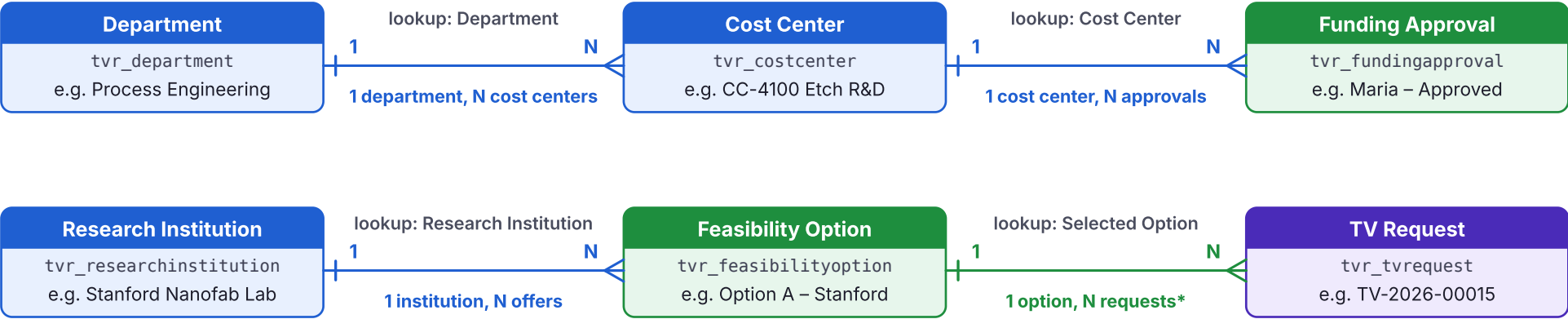


Self link "Previous Request":
a Repeat request points to the old one,
e.g. TV-2026-00016 → TV-2026-00003
(empty for New requests)

Picture 2 – The other links (12 links)

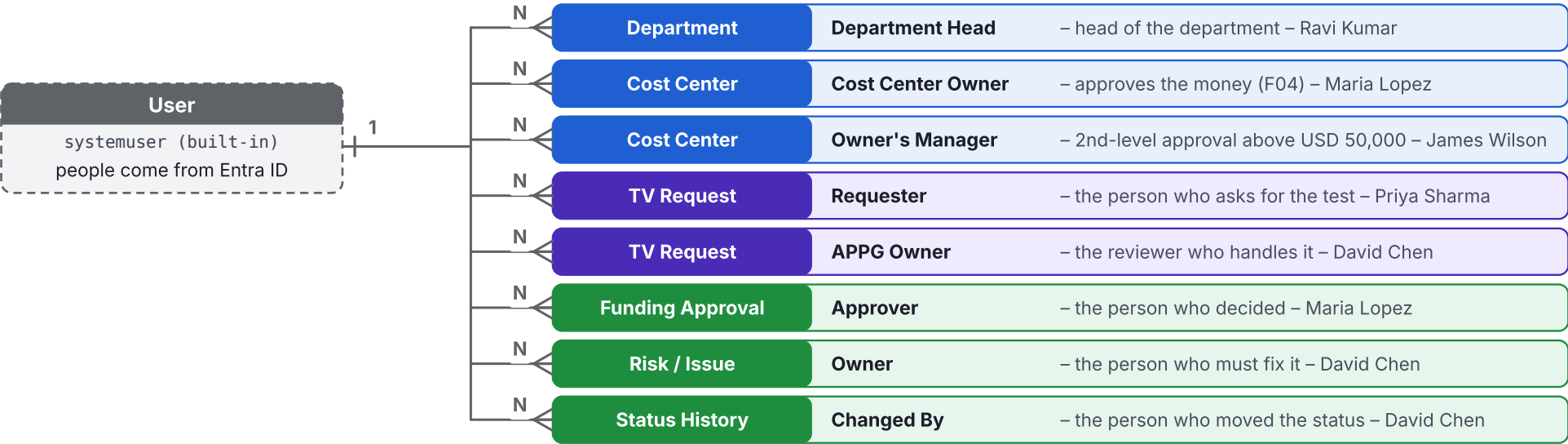
Same symbols as Picture 1. The small grey text on a line is the lookup column; it is stored on the N table.

A. THE OTHER 4 TABLE LINKS (not drawn in Picture 1)



* In practice exactly one request uses its own chosen option. The lookup is still a normal one-to-many (1 : N) link because it is stored on TV Request.

B. PERSON COLUMNS → USER (8 links) – one person can appear on many rows



1.4 Every link explained (26 links)

A. Links from master data (4 links)

#	ONE (1)	MANY (N)	Lookup column (on the N table)	What it means	Example	Picture
1	Department	TV Request	Department (<code>tvr_department</code>)	Every request belongs to one department; a department has many requests	TV-2026-00015 → Process Engineering	1
2	Location	TV Request	Location (<code>tvr_location</code>)	Where the request comes from	TV-2026-00015 → Fremont, CA	1
3	Cost Center	TV Request	Cost Center (<code>tvr_costcenter</code>)	The budget the request is charged to – its owner approves the money	TV-2026-00015 → CC-4100 Etch R&D	1
4	Department	Cost Center	Department (on Cost Center)	Each cost center belongs to one department	CC-4100 → Process Engineering	2

B. One request and its child tables (10 links). Each child table has a lookup column **TV Request** that points back to the request. The last column shows request TV-2026-00015 **at the end of the doc 13 story** (closed 5-Mar-2027).

#	ONE (1)	MANY (N)	What one child row means	TV-2026-00015 has ...	Picture
5	TV Request	Feasibility Option	One possible way to do the test (lab, cost, time)	3 rows: A Stanford 39,500 · B Fraunhofer 35,000 · C Internal Lab 22,000	1
6	TV Request	Funding Approval	One approval decision (one per approval level)	1 row: Maria Lopez – Approved – 39,500	1
7	TV Request	Milestone	One step of the work with planned and actual dates	4 rows: Wafer preparation, Etch runs, CD-SEM, Final report	1
8	TV Request	Risk / Issue	One problem or risk during execution	1 row: Lab tool down – High – Resolved	1
9	TV Request	Cost Entry	One real cost (invoice)	4 rows = 37,800, the request's <i>Actual Total Cost</i> (the sample CSV is an earlier moment: 2 rows = 23,300)	1
10	TV Request	Request Comment	One comment or question	2 rows: David asks for the wafer map, Priya answers	1
11	TV Request	Request Document	One uploaded or generated file (the file itself is in SharePoint)	7 rows in the doc 04 §5 list: spec, wafer map, quote, funding approval PDF, progress report, final report, summary PDF (more if all 3 quotes are uploaded)	1
12	TV Request	Status History	One status move (who, when, from → to)	11 rows: Draft → Submitted ... → Closed	1
13	TV Request	Notification Log	One e-mail the system tried to send (Sent or Failed)	One row per e-mail, e.g. REQ_SUBMITTED to Priya	1
14	TV Request	Error Log	One failed flow run	0 rows in this story (section 5.10 shows what a row looks like when a flow fails)	1

C. Other links between tables (4 links)

#	ONE (1)	MANY (N)	Lookup column	What it means	Example	Picture
15	Research Institution	Feasibility Option	Research Institution	Which lab or vendor made this offer. One lab can make offers for many requests	Option A → Stanford Nanofab Lab	2
16	Cost Center	Funding Approval	Cost Center	Which budget the approval is for	Maria's approval → CC-4100	2
17	TV Request (old)	TV Request (new)	Previous Request (<code>tvr_parentrequest</code>)	Self link. A <i>Repeat</i> request points to the request it repeats. Empty for <i>New</i> requests	TV-2026-00016 → TV-2026-00003	1 (dashed loop)
18	Feasibility Option	TV Request	Selected Option (<code>tvr_selectedoption</code>)	The one option chosen in Internal Alignment 2. It points to one of the request's own options (link 5)	TV-2026-00015 → Option A – Stanford	2

D. Person columns → User (8 links). In all of them **one** person can appear on **many** rows. All are in Picture 2, part B.

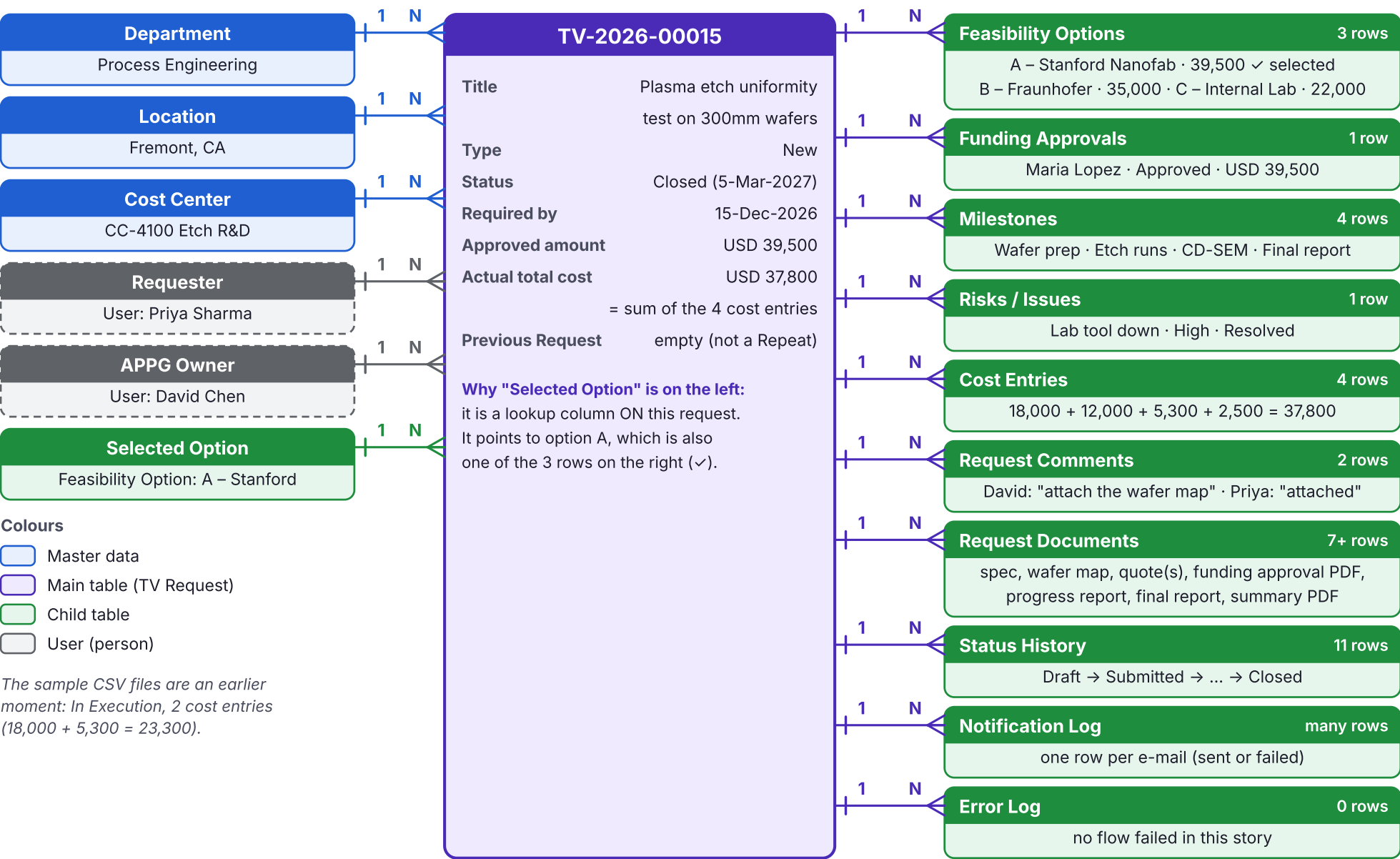
#	Table (N side)	Lookup column → User	Who is it?	Example
19	Department	Department Head	Head of the department	Ravi Kumar
20	Cost Center	Cost Center Owner	Approves the money (funding flow F04)	Maria Lopez
21	Cost Center	Owner's Manager	2nd-level approval for amounts above USD 50,000 (flow F04)	James Wilson
22	TV Request	Requester	The person who asks for the test	Priya Sharma
23	TV Request	APPG Owner	The reviewer who handles the request	David Chen
24	Funding Approval	Approver	The person who decided	Maria Lopez
25	Risk / Issue	Owner	The person who must fix the issue	David Chen
26	Status History	Changed By	The person who moved the status	David Chen

1.5 The same links with real data (Picture 3)

Picture 3 shows request **TV-2026-00015** at the end of the end-to-end use case (doc 13) and the real rows linked to it. **Left** = the one row this request points to in each parent table. **Right** = the many rows that point to this request. Note: the sample CSV files show the same request at an **earlier moment** (status *In Execution*, 2 cost entries = 23,300).

Picture 3 – Real data: request TV-2026-00015 (end of the doc 13 story, closed 5-Mar-2027)

Left: the ONE row this request points to in each parent table. Right: the MANY rows that point to this request.



1.6 How to create a link in Dataverse (step by step)

Example: link **TV Request** to **Department** (link 1).

1. Go to **make.powerapps.com** → environment **LAM-TV RMS - DEV** → **Solutions** → **TVRMS** → **Tables**.
2. Open the table on the **N side** – here **TV Request** (golden rule from 1.2).
3. Click **+ New** → **Column**. Display name: **Department**. Data type: **Lookup**. Related table: **Department**.
4. Click **Save**. Dataverse now creates the one-to-many relationship (one Department, many TV Requests) by itself.
5. Check it: open the **Department** table → **Relationships**. You see a *One-to-many* relationship to TV Request.
6. **Only for the 10 child-table links (table B):** set the behaviour from section 6 (Delete: Restrict, Assign/Share: Cascade All). For all other links (tables A, C and D) keep the **default** behaviour (no cascade). Otherwise, for example, re-assigning one Department row would also re-assign all requests of that department.

Create the lookup the same way for every row of tables A, B, C and D above (for the person columns, the related table is **User**). Common mistakes to avoid:

- Creating the lookup on the wrong table (on Department instead of TV Request). Then one department could have only one request.
- Using a text column (typing "Process Engineering") instead of a lookup. Then there is no real link, no dropdown and no report filtering.
- Creating a *many-to-many* relationship. None of the 26 links in this project needs many-to-many.

2. Master data tables (customer provides data – slide 8)

2.1 Department (**tvr_department**)

Column	Type	Required	Sample
Name (primary)	Text 100	Yes	Process Engineering
Department Code	Text 20	Yes	PE
Department Head	Lookup → User	No	Ravi Kumar
Is Active	Yes/No	Yes	Yes

2.2 Cost Center (**tvr_costcenter**)

Column	Type	Required	Sample
Name (primary)	Text 100	Yes	CC-4100 Etch R&D
Cost Center Code	Text 20	Yes	4100
Cost Center Owner	Lookup → User	Yes	Maria Lopez
Owner's Manager (for escalation / high value)	Lookup → User	No	James Wilson
Department	Lookup → Department	No	Process Engineering
Is Active	Yes/No	Yes	Yes

2.3 Location (`tvr_location`)

Column	Type	Sample
Name	Text	Fremont, CA
Country	Text	USA
Is Active	Yes/No	Yes

2.4 Research Institution / Vendor (`tvr_researchinstitution`)

Column	Type	Sample
Name	Text	Stanford Nanofab Lab
Type	Choice: University / Research Lab / Vendor / Internal Lab	University
Contact Name	Text	Dr. Alan Green
Contact Email	Email	agreen@stanford.example
Capabilities	Multiline text	Plasma etch, SEM, TEM
Is Active	Yes/No	Yes

2.5 App Setting (`tvr_appsetting`) – business can change rules without code

Name (Key)	Value	Description
ReminderAfterDays	3	Send reminder if pending action > 3 days
EscalationAfterDays	5	Escalate to manager if pending > 5 days
AutoCancelAfterDays	30	Auto-cancel if "More Info Required" > 30 days
HighValueFundingLimit	50000	Above this, 2nd-level approval
RetentionYears	7	Record retention

2.6 Email Template (`tvr_emailtemplate`)

Column	Type	Sample
Template Code (primary)	Text	REQ_SUBMITTED
Subject	Text	[TVRMS] {TVID} submitted – {Title}
Body (HTML)	Multiline	Dear {RequesterName}, Your request {TVID} has been submitted...
Is Active	Yes/No	Yes

The flow replaces `{TVID}`, `{Title}`, `{RequesterName}`, `{Status}`, `{Link}`, `{Comments}` with real values.

3. Main table – TV Request (`tvr_tvrequest`)

Turn on: **Auditing, Track changes**. Ownership: **User or team**.

#	Display name	Schema name	Type	Req.?	Sample value
1	TVID (primary)	<code>tvr_tvid</code>	Autonumber TV- {DATETIMEUTC:yyyy} - {SEQNUM:5}	auto	TV-2026-00015
2	Request Title	<code>tvr_title</code>	Text 200	Yes	Plasma etch uniformity test on 300mm wafers
3	Request Type	<code>tvr_requesttype</code>	Choice: New, Repeat, One-time, Recurring	Yes	New
4	Previous Request (for Repeat)	<code>tvr_parentrequest</code>	Lookup → TV Request	If Repeat	TV-2026-00003
5	Recurrence Frequency	<code>tvr_frequency</code>	Choice: Monthly, Quarterly, Half- yearly, Yearly	If Recurring	Quarterly
6	Requester	<code>tvr_requester</code>	Lookup → User	Yes	Priya Sharma
7	Department	<code>tvr_department</code>	Lookup → Department	Yes	Process Engineering
8	Location	<code>tvr_location</code>	Lookup → Location	Yes	Fremont, CA
9	Cost Center	<code>tvr_costcenter</code>	Lookup → Cost Center	Yes	CC-4100 Etch R&D
10	Business Justification	<code>tvr_justification</code>	Multiline 4000	Yes	Customer X reports 4% non-uniformity...
11	Technical Description	<code>tvr_technicaldescription</code>	Multiline 4000	Yes	25 wafers, 3 recipes, measure CD...
12	Expected Outcome	<code>tvr_expectedoutcome</code>	Multiline	Yes	Uniformity < 2%
13	Priority	<code>tvr_priority</code>	Choice: Low, Medium, High, Critical	Yes	High
14	Required By Date	<code>tvr_requiredbydate</code>	Date only	Yes	2026-12-15
15	Estimated Budget (USD)	<code>tvr_estimatedbudget</code>	Currency	Yes	42,000
16	Request Status (do NOT name it just "Status" – Dataverse already has a system column called Status)	<code>tvr_status</code>	Choice (see §4)	auto	Submitted
17	Current Phase	<code>tvr_phase</code>	Choice: Initiation, APPG Review, Internal Alignment, Funding Approval, Execution, Closure	auto	APPG Review
18	APPG Owner	<code>tvr_appgowner</code>	Lookup → User	at review	David Chen
19	Selected Option	<code>tvr_selectedoption</code>	Lookup → Feasibility Option	at alignment 2	Stanford Nanofab – Option A

#	Display name	Schema name	Type	Req.?	Sample value
20	Approved Amount (USD)	tvr_approvedamount	Currency	at funding	39,500
21	Expected Completion Date	tvr_expectedcompletion	Date only	execution	2027-02-28
22	Overall Progress %	tvr_progress	Whole number 0-100	execution	60
23	Actual Total Cost (USD)	tvr_actualcost	Currency (Rollup of Cost Entry)	auto	37,800
24	SharePoint Folder URL	tvr_spfolderurl	URL	auto	https://lam.sharepoint.com/sites/TVRMS/TV Requests/TV-2026-00015
25	Submitted On	tvr_submittedon	Date and time	auto	2026-10-12 10:05
26	Status Changed On	tvr_statuschangedon	Date and time	auto	used for reminders/aging
27	Closed On	tvr_closedon	Date and time	auto	2027-03-05
28	Final Outcome	tvr_finaloutcome	Multiline	at closure	Uniformity improved to 1.6%
29	Lessons Learned	tvr_lessonslearned	Multiline	at closure	Book lab 6 weeks early
30	Reuse Recommendation	tvr_reuserecommendation	Choice: Reuse as-is, Reuse with changes, Do not reuse	at closure	Reuse with changes
31	Rejection / Cancellation Reason	tvr_reason	Multiline	when rejected/cancelled	Duplicate of TV-2026-00011
32	Cycle Time (days)	tvr_cycletimedays	Formula/Calculated: DiffInDays (Submitted0n, Closed0n)	auto	144

4. Status list and allowed moves (ASSUMPTION – confirm in workshop)

Code	Status	Phase	Who moves it next	Allowed next status
1	Draft	Initiation	Requester	Submitted, Cancelled
2	Submitted	APPG Review	APPG	Under APPG Review
3	Under APPG Review	APPG Review	APPG	More Info Required, Rejected, Internal Alignment 1
4	More Info Required	APPG Review	Requester	Under APPG Review (resubmit), Cancelled (auto after 30 days)
5	Rejected	APPG Review	– (final)	–
6	Internal Alignment 1	Internal Alignment	APPG	Internal Alignment 2
7	Internal Alignment 2	Internal Alignment	APPG	Pending Funding Approval
8	Pending Funding Approval	Funding Approval	CCO (via flow)	Funding Approved, Funding Rejected

Code	Status	Phase	Who moves it next	Allowed next status
9	Funding Rejected	Funding Approval	– (final)	–
10	Funding Approved	Funding Approval	APPG	In Execution
11	In Execution	Execution	APPG	On Hold, Execution Completed
12	On Hold	Execution	APPG	In Execution, Cancelled
13	Execution Completed	Closure	APPG	Closed
14	Closed	Closure	– (final)	–
15	Cancelled	any	– (final)	–

Tip: put the "allowed next status" rules in **one place** (the Canvas app uses a collection or a small table `tvr_statustransition`) so the rules are easy to change.

5. Child tables

5.1 Feasibility Option (`tvr_feasibilityoption`)

Column	Type	Sample
Option Name (primary)	Text	Option A – Stanford Nanofab
TV Request	Lookup → TV Request	TV-2026-00015
Research Institution	Lookup	Stanford Nanofab Lab
Capability Score (1-5)	Whole number	5
Quoted Cost (USD)	Currency	39,500
Timeline (weeks)	Whole number	10
Constraints	Multiline	Lab available only from Nov
Recommendation	Choice: Recommended, Alternative, Not Recommended	Recommended
Is Selected	Yes/No	Yes
Quote Document URL	URL	link to SharePoint file

5.2 Funding Approval (`tvr_fundingapproval`)

Column	Type	Sample
Name	Text (auto)	FA-TV-2026-00015-1
TV Request	Lookup	TV-2026-00015
Cost Center	Lookup	CC-4100
Approver	Lookup → User	Maria Lopez
Approval Level	Whole number	1
Requested Amount	Currency	39,500
Decision	Choice: Pending, Approved, Rejected	Approved
Comments	Multiline	Approved from Q4 budget
Requested On / Decided On	Date time	2026-11-02 / 2026-11-03

This table **is** the "approval history" asked in scope.

5.3 Milestone (`tvr_milestone`)

Column	Type	Sample
Milestone Name	Text	Wafer preparation
TV Request	Lookup	TV-2026-00015
Sequence	Whole number	1
Planned Start / Planned End	Date	2026-11-10 / 2026-11-24
Actual Start / Actual End	Date	2026-11-12 / 2026-11-26
% Complete	Whole number	100
Status	Choice: Not Started, In Progress, Completed, Delayed	Completed

5.4 Risk / Issue (`tvr_riskissue`)

Column	Type	Sample
Title	Text	Lab tool down for maintenance
TV Request	Lookup	TV-2026-00015
Type	Choice: Risk, Issue, Exception	Issue
Severity	Choice: Low, Medium, High	High
Owner	Lookup → User	David Chen
Mitigation / Action	Multiline	Move run to 2nd lab
Status	Choice: Open, In Progress, Resolved	Resolved
Due Date	Date	2026-12-05

5.5 Cost Entry (`tvr_costentry`)

Column	Type	Sample
Description	Text	Lab usage fee – Nov
TV Request	Lookup	TV-2026-00015
Category	Choice: Lab Fee, Material, Shipping, Labor, Other	Lab Fee
Amount (USD)	Currency	18,000
Cost Date	Date	2026-11-30
Invoice Reference	Text	INV-88213

Create a **Rollup column** on TV Request: `Actual Total Cost = SUM(Cost Entry.Amount)` . **Utilization %** = Actual Total Cost / Approved Amount × 100 (calculated in Power BI or formula column).

5.6 Request Comment (`tvr_requestcomment`)

Column	Type	Sample
Comment	Multiline	Please attach the wafer map.
TV Request	Lookup	TV-2026-00015
Stage	Choice (same as Phase)	APPG Review
Comment Type	Choice: General, More Info Request, Requester Response, Decision	More Info Request
Created By / Created On	system	David Chen / 2026-10-13

5.7 Request Document (`tvr_requestdocument`) – metadata only, file lives in SharePoint

Column	Type	Sample
File Name	Text	Etch_Test_Spec_v1.pdf
TV Request	Lookup	TV-2026-00015
Document Type	Choice: Technical Specification, Supporting Document, Quote, Approval Evidence, Progress Report, Final Report	Technical Specification
SharePoint URL	URL	...
Folder	Text	02-Technical Documents
Uploaded On	Date time	2026-10-12

5.8 Status History (`tvr_statushistory`) – audit trail in readable form

Column	Type	Sample
Name	Text	TV-2026-00015: Submitted → Under APPG Review
TV Request	Lookup	TV-2026-00015
From Status / To Status	Choice	Submitted / Under APPG Review
Changed By	Lookup → User	David Chen
Changed On	Date time	2026-10-13 09:12
Remarks	Multiline	Picked up for review

5.9 Notification Log (`tvr_notificationlog`)

Column	Type	Sample
Subject	Text	[TVRMS] TV-2026-00015 submitted
TV Request	Lookup	TV-2026-00015
Template Code	Text	REQ_SUBMITTED
Sent To	Text	priya.sharma@lam.com
Sent On	Date time	2026-10-12 10:06
Result	Choice: Sent, Failed	Sent
Error Message	Multiline	(empty)

5.10 Error Log (`tvr_errorlog`) – for "Exception Remediation"

Column	Type	Sample
Flow Name	Text	F01 - On Request Submitted
TV Request	Lookup	TV-2026-00015
Error	Multiline	SharePoint: folder already exists
Run URL	URL	link to failed flow run
Status	Choice: New, Fixed, Ignored	New

6. Relationships – important settings

For all child tables → TV Request lookups, set relationship behavior:

- **Delete: Restrict** (never lose history; 7-year retention).
- **Assign/Share: Cascade All** (so when the request is shared with a CCO or stakeholder, children are shared too).

7. Business rules / validations in Dataverse (server side)

Even if the Canvas app checks fields, also add these (so data is safe from any entry point):

1. **Business Rule** on TV Request: if Request Type = Repeat → *Previous Request* is required.
2. **Business Rule**: if Request Type = Recurring → *Recurrence Frequency* is required.
3. **Business Rule**: Required By Date must be after today (show error).
4. Column *Estimated Budget* min value = 0.

8. How to load sample data

Files in `/sample-data`. Load in this order (because of lookups):

1. `departments.csv`
2. `locations.csv`
3. `research_institutions.csv`
4. `cost_centers.csv`
5. `app_settings.csv`
6. `email_templates.csv`
7. `tv_requests.csv`
8. `feasibility_options.csv`
9. `milestones.csv`
10. `cost_entries.csv`

How: open the table → **Import** → **Import data from Excel/CSV** → map columns → Import. (Users must exist in the environment first – they come from Entra ID.)